Type MS · Express a part as a percentage of a total

Answer 1 of 7 · Question 7 part b.ii · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
7	(a)	Any three from: • changes the DNA/base/codon sequence; • changes the order of the amino acids/different amino acid coded for; • changes the function/shape of the protein/different protein; • affects a characteristic or function/genotype in the individual;		3
	(b) (i)	Two from: • dialysis has to be completed several times a week/transplant permanent; • increased life expectancy; • less restrictions on diet; • better quality of life/freedom to move around/normal lifestyle;		2
	(ii)	100 x 3233÷4733; = 68.3/68		2
	(iii)	Two from: • people have two kidneys; • two recipients from one donor; • living donation/can live with just one;		2
	(c) (i)	0.5:1;	Accept 1:2	1
	(ii)	individual B: d, d individual A: D → Dd, Dd; d → dd, dd • correct genotype individual A; • correct genotype individual B; • 50% offspring heterozygous/correct genotype and affected; • 50% homozygous recessive/correct genotype and not affected;		4

Total 14 marks

Type MS · Express a part as a percentage of a total

Answer 1 of 7 · Question 7 part b.ii · 2 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

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Type MS · Express a part as a percentage of a total

Answer 2 of 7 · Question 4 part a.iv · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a.iv in this mark scheme extract

4(a)(i)	A protein;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	A the molecules are too large;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(a)(iii)	• more glucose in blood/filtrate; • as little/no insulin to lower blood glucose levels/glucose not converted to glycogen; • not all/too much to be reabsorbed; • some found in urine;		**(3)**

Question Number	Answer	Additionalguidance	Mark
4(a)(iv)	$\frac{25 \times 100}{53}$; 47%;	Full marks for correct final answer Allow 47.2% / 47.17 Ecf allow one mark for correct answer using incorrect figures in calculation	**(2)**

Type MS · Express a part as a percentage of a total

Answer 2 of 7 · Question 4 part a.iv · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a.iv in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
4(b)	Any five from: Advantages • transplant permanent/carried out once; • can return to normal life; • not connected to machine for long periods/no need for frequent hospital visits; Disadvantages • risk of rejection; • risk of infection (during/after surgery)/reduced immunity; • difficulties in finding donor/long waiting lists for donor;		**(5 max)**

Total 12 Marks

Type MS · Express a part as a percentage of a total

Answer 2 of 7 · Question 4 part a.iv · 2 marks

2019 JUN · Paper 01 (regional R) · extract 3 of 3 · find part a.iv in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Express a part as a percentage of a total

Answer 3 of 7 · Question 3 part a.ii · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

3(a)(i)	B 3.2 - 3.5 kg;		**(1)**

Question Number	Answer	Additional guidance	Mark
3(a)(ii)	(94 / 210) x 100 ; 45%; (2)	Ecf allow correct answer to incorrect figures used in calculation Allow 44.76%/ 44.8% for correct final answer	**(2)**

Question Number	Answer	Additional guidance	Mark
3(b)	smoking appears to lower birth mass; more higher birth mass babies born to non-smokers;	ORA for both mp's	**(2)**

Question Number	Answer	Additional guidance	Mark
3(c)	• cigarette smoke contains CO; • combines with haemoglobin instead of oxygen/forms carboxyhaemoglobin; • less oxygen to fetus; • less respiration in fetus; • lower growth rate/cell division; • causes lower birth mass;		**(5 max)**

Total 10 Marks

Type MS · Express a part as a percentage of a total

Answer 3 of 7 · Question 3 part a.ii · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Express a part as a percentage of a total

Answer 4 of 7 · Question 4 part b.iii · 2 marks

2020 NOV · Paper 01 (regional R) · find part b.iii in this mark scheme extract

4 (a)	• ankles(1) • protein(1)		2
(b) (i)	• axes labels(1) • axes scales(1) • correctly plotted bars(1)	Max 2 marks for line graph	3
(ii)	improved/varied/balanced diet/education (in nutrition) / international aid/less protein in diet in 2010(1)		1
(iii)	• 95 ÷ 1978 (x 100) (1) • 4.8(%)	Full marks for correct final answer	2

Total for Question 4 = 8 marks

Type MS · Express a part as a percentage of a total

Answer 5 of 7 · Question 4 part a.ii · 3 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

4 (a) (i)	• suitable scales; • axes labelled; • correct plots;;	Number of teeth on X axis, number of students on Y axis	4
(ii)	• 13; / 30 • x 100; • 43.3(%);	Allow 43(%)	3
(b)	• wisdom teeth/molars not developed; • jaw too small to accommodate extra teeth;		2
(c)(i)	• sugar coats/left in teeth/forms plaque; • bacteria use sugar (for respiration/energy); • convert sugar to lactic acid; • acid dissolves/erodes/breaks down enamel;	Allow feed on Allow release acid	4
(c)(ii)	any two from • regular brushing of teeth/use fluoride toothpaste/flossing; • visit dentist regularly; • add fluoride to tap water		2 Total 15

Type MS · Express a part as a percentage of a total

Answer 5 of 7 · Question 4 part a.ii · 3 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Express a part as a percentage of a total

Answer 6 of 7 · Question 4 part a.ii · 2 marks

2024 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

4 (a)	(i)	11, 31, 30		(1)
	(ii)	31 (× 100); 100 = 31%;	send to review if other than 31	(2)
(b)	(i)	**D (alternative forms of the same gene in a genotype)** The following are incorrect because: A two dominant genes are the same type of allele B two recessive genes are the same type of allele C alleles are not different genes		(1)
	(ii)	• sperm and egg/gametes each contain one allele/haploid; • sperm and egg/gametes fuse; • during fertilisation; • (fertilisation) brings alleles together (in diploid cell)/chromosomes (from each gamete) are paired/one allele comes from each parent;		(4)
(c)	(i)	Hh/heterozygous; hh/homozygous recessive;	either order any suitable letters to be used for alleles	(2)
	(ii)	wheezing/coughing/shortness of breath/low weight / fatigue/over production of mucus/diarrhoea/male infertility;	allow other valid symptoms	(1)

Total for Question 4 = 11 marks

Type MS · Express a part as a percentage of a total

Answer 7 of 7 · Question 8 part b.i · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

8 (a)	Three from: • (parasite) carried by mosquito/mosquito is a vector; • mosquito bites human; • parasite/infected blood drawn/sucked up into mosquito; • infected blood transmitted to other people;		Max 3
(b)(i)	• fewer deaths caused by *P.vivax* in R than Q • (but) more deaths caused by *P.vivax* in R than in Q as a proportion of the total; • 190 000 fewer deaths caused by *P.vivax* in R/300 less deaths in Q caused by *P.vivax*; • 0.5% deaths caused by *P.vivax* in Q/62.5% of deaths caused by *P.vivax* in R;		1 1 1
(ii)	more mosquitoes in one region than another/climate favours breeding of mosquitoes/more dense population of people/better health care/use of (named) preventative measures;		1
(c)	Three from: Fewer people with malaria/reduced incidence of malaria; Immunity/antibodies against parasite/herd immunity; Reduction in population of/death of parasite; Less transmission (from one person to another);		Max 3
(d)	Three from: (sexual reproduction) • gives rise to variation in offspring; • variation provides a survival advantage; • parasite more likely to survive in a changing environment; (asexual reproduction) • parasite can reproduce faster; • no need to find a mate/only one parent needed; • larger number of offspring produced;	Answer must contain at least one advantage of each method	Max 3

Type MS · Express a part as a percentage of a total

Answer 7 of 7 · Question 8 part b.i · 3 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks